

Mechanical Engineering Portfolio

Ruitao Zhao

Mechanical Engineering | Robotics & Manufacturing

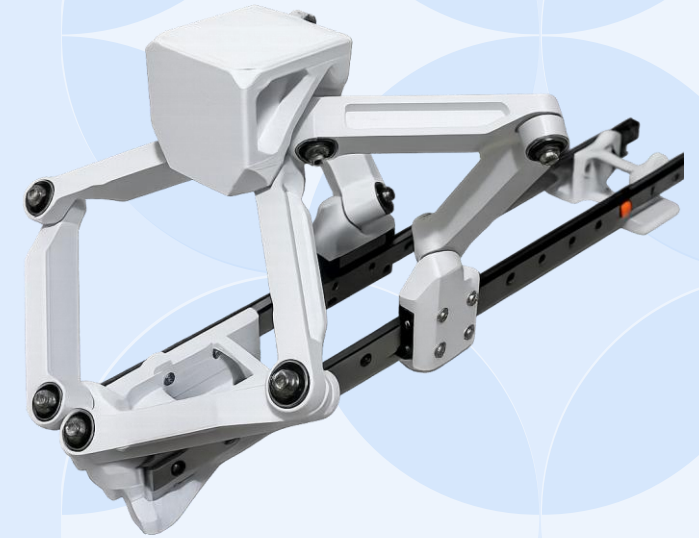
NORTHEASTERN - SELECTED PROJECTS

Parallel Mechanism Reconstruction & Prototyping

- Reconstructed a 3-RPS parallel wrist mechanism from reference models by simplifying kinematic representations.
- Enabled workspace and singularity evaluation through MATLAB-based modeling.
- Examined stiffness–workspace trade-offs across simplified kinematic representations.
- Checked the consistency between simplified models and physical behavior using 3D-printed prototypes.

ROS2 GPS Sensing & Coordinate Transformation

- Implemented a ROS2 (Humble) sensing pipeline in Python (rclpy) for serial-based GPS data acquisition.
- Developed custom ROS2 nodes for NMEA (GPGGA) parsing and LL-to-UTM coordinate transformation.
- Compared positioning performance across open, occluded, and dynamic environments using ROS2 log data and MATLAB.



3-RPS parallel wrist mechanism 3D-printed prototype

SIASUN ROBOTICS

Scope of Responsibility

- Supported manufacturing and production engineering for multiple industrial robot models, with primary responsibility for the SR12A series
- Acted as the engineering interface between mechanical design intent and production execution

Core Engineering Contributions

- Translated design-stage BOMs and assembly concepts into production-ready engineering BOMs and SOPs
- Defined part hierarchy, configuration rules, and assembly logic to support stable mass production
- Standardized assembly workflows to ensure consistency across production builds



Primary production-supported model: SR12A

SIASUN ROBOTICS

Production Optimization & Issue Handling

- Supported continuous production optimization by refining BOM structures and assembly SOPs
- Identified process ambiguities and updated documentation to reduce shop-floor variation
- Analyzed production and batch-level issues, proposing corrective actions for subsequent builds

Skills & Engineering Impact

- Engineering BOM and configuration management
- Design-to-manufacturing transition for industrial robots
- Production SOP development and optimization
- Manufacturing issue analysis and cross-functional collaboration



Additional production-supported model: SR12C

YINGZHONG TECHNOLOGY

Lifting Mechanism Design & Prototyping

- *(Product Engineering & Mechanical Integration)*
- Designed a synchronized lifting mechanism in SolidWorks to support large-format panels while maintaining balanced motion and preventing mechanical interference.
- Developed and refined sheet-metal structures and surface-finishing specifications through on-site engineering practice and supplier coordination.
- Introduced a simplified positioning jig to improve assembly precision and repeatability during production.
- Supported mechanical integration from initial concept to first-generation prototype validation.
- A lifting synchronization mechanism, ZL202323350721
- A lifting mechanism and adjustable desk, ZL02322302202



First-generation prototype of a lifting TV cabinet mechanism

Thank you

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